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Are Patents with Female Inventors Under-Cited? Evidence from Text Estimation

Contents

1 Big picture

- **Research question.** Are patents with female lead inventors under-cited relative to what they would have received if the same patent had a male lead inventor?
- **Main idea.** Forward citations are widely used as a proxy for patent quality. The paper asks whether citation counts also reflect gender-based recognition frictions, so that equal-quality patents may receive different citation credit depending on the gender of the lead inventor.
- **Empirical object.** The paper estimates counterfactual forward citations using patent text, patent novelty, and patent impact. The key comparison is the actual citation count for a patent versus the citation count predicted for the same patent under a male lead-inventor benchmark.
- **Methodological innovation.** The authors introduce **Inference from Text Analysis**, or **I-TEXT**. I-TEXT uses transformer text embeddings, a gender propensity model, and separate citation prediction networks for male and female lead-inventor patents.
- **Main finding.** Patents with female lead inventors receive fewer citations than the same patents are predicted to receive if the lead inventor were male. In the main regression-adjusted estimates, the gap is about 2.5 citations per patent.
- **Who contributes to the gap.** The largest contribution comes from citations added by male lead inventors in future patent applications. Female lead inventors also undercite female-led patents to a smaller degree. Examiner-added citation gaps are much smaller, and female examiners appear closest to even-handed.
- **Market-value result.** Using market-based patent values from Kogan et al. (2017), the paper finds no economically meaningful gender difference in patent value after controlling for patent characteristics and text-based quality measures.
- **Economic takeaway.** Citation counts are not a neutral measure of patent quality. If patents with female inventors are systematically under-recognized in citations, research and career evaluations based on patent citations may understate female inventors' contributions.

2 Incremental contribution of the paper

- **From citation counts to citation bias.** The paper moves beyond documenting that female inventors receive fewer citations. It asks whether the lower citation count is justified by lower patent quality or instead reflects under-recognition.
- **Text-based quality adjustment.** Existing patent-quality measures often rely on citations themselves. This paper instead uses the patent's textual content, novelty, and impact to estimate expected citations, which helps separate quality from realized recognition.
- **New method for economics and finance.** The paper introduces I-TEXT as a framework for using high-dimensional text to estimate group gaps in outcomes. This is a methodological contribution for empirical work where the object being evaluated is itself a text document.
- **Counterfactual comparison at the patent level.** Rather than comparing average male and female patents only with observable controls, I-TEXT estimates how many citations the same patent would have received under a different lead-inventor gender.

- **Decomposition of citation sources.** The paper separates inventor-added and examiner-added citations, and further decomposes them by the gender of the inventor or examiner adding the citation.
- **Heterogeneity across fields and time.** The analysis shows that the undercitation pattern is broad across technology areas, stronger in some emerging fields, and persistent over the sample period.
- **Market validation.** The finding that market values do not differ by lead inventor gender supports the interpretation that the citation gap is not simply lower economic value of female-led patents.
- **Implication for prior literature.** Since patent citations are a common proxy for innovation quality, gender-related citation bias can affect many empirical studies using forward citations as either an outcome or a control.

3 Data and measurement

3.1 Patent sample

The main patent data come from the USPTO. The core sample covers U.S. utility patents granted from 1976 through 2015. The main analysis focuses on patents that receive at least one forward citation, because the modal patent receives zero citations.

- **Unit of observation.** A granted utility patent.
- **Main gender measure.** The gender of the first-listed inventor, referred to as the lead inventor.
- **Main citation measure.** The number of forward citations received by the patent.
- **Main estimation sample.** Table 1 reports 521,967 patents with positive citation counts: 289,847 male lead-inventor patents and 232,120 female lead-inventor patents.
- **Technology controls.** The paper uses art units, CPC classifications, and NBER patent categories to capture technology fields.

Interpretation. The sample is designed to compare patents that are sufficiently visible to receive at least one citation, and then ask whether female-led patents receive fewer citations than their content and context would predict.

3.2 Inventor gender and citation sources

- **Lead inventor gender.** Inventor gender comes from PatentView. The baseline definition uses the first-listed inventor, but robustness checks use single-inventor patents and team-gender definitions.
- **Inventor-added citations.** Patent applicants cite prior patents in their applications.
- **Examiner-added citations.** Patent examiners may add citations that they view as missing from the application.

- **Citation-source availability.** The USPTO begins identifying examiner-added citations in the early 2000s, so source-decomposition tests use a later subsample.
- **Examiner gender.** Examiner gender is inferred from first names using Social Security Administration and WIPO name-frequency data.

Interpretation. Separating inventor-added and examiner-added citations matters because undercitation can arise either from future inventors not recognizing prior female-led patents or from examiners failing to add them.

3.3 Patent characteristics and fixed effects

The paper uses detailed patent-level controls and fixed effects to account for non-quality citation determinants.

- **Art unit fixed effects** control for narrow technology areas inside the USPTO.
- **Patent issue-year fixed effects** control for time trends and citation opportunities.
- **Examiner fixed effects** control for persistent citation behavior of examiners.
- **Attorney fixed effects** control for common writing and filing practices by patent attorneys.
- **Assignee fixed effects** control for firm-level patenting and citation environments.

Interpretation. The preferred specifications ask whether the gender gap remains after comparing patents within the same technology areas, time periods, examiner environments, attorney environments, and assignee firms.

3.4 Novelty, impact, and significance

The paper follows Kelly et al. (2021) to measure how important a patent is relative to the past and future patent corpus.

- **Forward similarity / impact.** A patent is more impactful if later patents use language similar to it. The paper measures forward similarity over the five years after the focal patent is granted:

$$F_i^5 = \frac{1}{|S_F^5|} \sum_{j \in S_F^5} \mathrm{sim}(i, j).$$

- **Backward similarity.** A patent is less novel if it is very similar to patents granted in the prior five years:

$$B_i^5 = \frac{1}{|S_B^5|} \sum_{j \in S_B^5} \mathrm{sim}(i, j).$$

- **Novelty.** Novelty is increasing when backward similarity is low:

$$\mathrm{Novelty}_i = \frac{1}{B_i^5}.$$

- **Significance.** Overall significance scales impact by backward similarity:

$$\text{Significance}_i = \frac{F_i^5}{B_i^5}.$$

Interpretation. A high-significance patent looks different from recent prior patents and looks similar to future patents. This helps distinguish patents that are merely in active fields from patents whose content is genuinely influential.

3.5 Text embeddings

The core text input is the full patent text from PatentView, excluding header information such as author, firm, and location.

- **Main encoder.** The baseline uses Longformer, a transformer architecture designed for long documents.
- **Embedding dimension.** The encoder produces 768-dimensional patent-text embeddings.
- **Robustness encoder.** The paper also uses SciBERT, trained on scientific writing, in robustness tests using patent abstracts.

Interpretation. The embeddings reduce the patent text to numerical features that retain semantic content. These features then enter the citation-prediction model.

4 Methodology and empirical specifications

4.1 Raw citation regression

The starting point is an OLS regression of forward citations on a female lead-inventor indicator and fixed effects:

$$Y_i = \beta_1 I(\text{FemaleInventor}_i) + \delta_{\text{GrantYear}} + \delta_{\text{ArtUnit}} + \delta_{\text{Examiner}} + \delta_{\text{Attorney}} + \delta_{\text{Assignee}} + \varepsilon_i.$$

Here, Y_i is the actual number of forward citations. A negative β_1 means female lead-inventor patents receive fewer realized citations than male lead-inventor patents, conditional on the included controls.

Interpretation. This regression documents the raw conditional citation gap, but it does not answer whether the gap reflects lower patent quality or under-recognition.

4.2 I-TEXT architecture

I-TEXT estimates counterfactual citations using patent text, novelty, and impact.

- **Step 1: encode the patent text.** Longformer maps the patent text W_i into an embedding vector.
- **Step 2: add contextual importance.** The text embedding is combined with novelty and impact to form Z_i.

- **Step 3: estimate gender propensity.** A propensity network estimates the probability that the patent has a female lead inventor:

$$\hat{g}(Z_i) = P(T_i = 1 \mid Z_i),$$

where $T_i = 1$ indicates a female lead inventor.

- **Step 4: estimate gender-specific citation mappings.** The model fits two citation networks:

$$\hat{Q}_1(Z_i) = E[Y_i(1) \mid Z_i], \qquad \hat{Q}_0(Z_i) = E[Y_i(0) \mid Z_i],$$

where $Y_i(1)$ is the citation outcome under a female lead inventor and $Y_i(0)$ is the citation outcome under a male lead inventor.

Common-support filter. Patents whose text makes the lead-inventor gender too predictable are dropped using the propensity model. This is intended to avoid counterfactual estimates for patents with no comparable opposite-gender support.

4.3 Counterfactual gender citation gap

The I-TEXT gap is the average difference between predicted citations under the female and male mappings:

$$\text{Gap} = \frac{1}{N}\sum_{i=1}^{N}\left[E(Y_i(1) \mid Z_i) - E(Y_i(0) \mid Z_i)\right] = \frac{1}{N}\sum_{i=1}^{N}\left[\hat{Q}_i(1, Z_i) - \hat{Q}_i(0, Z_i)\right].$$

A negative gap means that, holding patent content and context fixed, the model predicts fewer citations when the lead inventor is female than when the lead inventor is male.

Interpretation. The counterfactual comparison is not a simple comparison of different male and female patents. It asks how the same patent would be cited under different lead-inventor gender assignments.

4.4 Patent-level delta

The main regression outcome is the difference between actual citations and the I-TEXT estimate under a male lead-inventor benchmark:

$$\Delta_i = \text{ForwardCitation}_i - \widehat{\text{ForwardCitation}}_{i|T_i=0}.$$

- If $\Delta_i > 0$, the patent received more citations than predicted under the male benchmark.
- If $\Delta_i < 0$, the patent received fewer citations than predicted under the male benchmark.
- A negative coefficient on the female lead-inventor indicator in a Δ_i regression means female-led patents are under-cited relative to the male benchmark.

The main regression replaces actual citations with Δ_i as the dependent variable and adds the same fixed effects used in the raw citation regressions.

4.5 Heterogeneity and decomposition

The paper uses the I-TEXT framework for several additional tests.

- **Technology heterogeneity.** The authors estimate undercitation across NBER patent subcategories and CPC categories.
- **Emerging fields.** A patent is classified as being in an emerging field if its art unit first appeared within five years of the patent grant.
- **Time evolution.** The paper estimates the gender-based citation gap by grant year using forward citations within the first ten years after grant.
- **Source of citations.** The paper separates citations added by inventors from citations added by examiners, and splits both by gender.
- **Robustness.** Tests include different gender definitions, inclusion of zero-citation patents, alternative embeddings, self-citation exclusions, placebo assignment, writing-style controls, and alternative transformations of Δ.
- **Market value.** The paper uses market-based patent values to test whether female-led patents have lower economic value.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies a text-adjusted citation gap: the difference between realized citations and the citations predicted from patent content, novelty, and impact under a male lead-inventor benchmark. The strongest interpretation is that female-led patents receive less citation recognition than comparable male-led patents with similar textual and contextual quality.

5.2 Key I-TEXT assumptions

- **Quality is captured by text and context.** The patent text, novelty, and impact must contain enough information to measure patent quality relevant to citations.
- **Embeddings preserve meaningful content.** Longformer and SciBERT embeddings must capture the semantic information needed to predict citations and gender propensity.
- **Common support.** The method requires comparable male and female patents in the embedding space. The propensity filter is designed to remove patents whose gender is too predictable from text.
- **Fixed effects capture non-quality citation channels.** Art unit, year, examiner, attorney, and assignee fixed effects are used to absorb institutional and field-specific differences not captured by text.

5.3 Why the result is not just lower quality

The paper's central concern is that lower citations for female-led patents could reflect lower patent quality. The market-value tests address this concern. If female-led patents were truly lower value, one might expect public markets to assign lower values to them. Instead, after controls, market values are not meaningfully lower for female-led patents.

5.4 What the paper does not fully identify

The paper does not conclusively identify the motive behind undercitation. The pattern is consistent with bias or under-recognition, but it cannot distinguish cleanly among taste-based discrimination, statistical discrimination, network differences, visibility differences, or field-specific information-flow frictions.

5.5 Interpretation of Δ

Δ_i is not a structural parameter. It is a patent-level citation premium relative to the citation count predicted by I-TEXT under a male lead-inventor benchmark. A negative female coefficient in the Δ regression means female-led patents receive fewer realized citations than expected, conditional on the patent's textual and contextual features.

6 Tables and figures

6.1 Tables 1 and 2: Sample statistics and baseline undercitation

Read Table 1:

- Male lead-inventor patents receive an average of 23.62 forward citations, compared with 19.2 for female lead-inventor patents.
- The average standardized significance score is higher for male lead-inventor patents and lower for female lead-inventor patents.
- Top-decile innovations are more common among male lead-inventor patents in the sample: 11% versus 8%.
- CPC distributions differ by lead-inventor gender, which motivates technology fixed effects and field heterogeneity tests.

Read Table 2:

- Panel A shows that female lead-inventor patents receive fewer actual forward citations in raw fixed-effect regressions.
- Panel B replaces actual citations with Δ, the text-adjusted citation premium relative to the male benchmark.
- The female lead-inventor coefficient is consistently around -2.25 to -2.47 citations in the text-adjusted specifications.

- The estimate remains stable after adding art group, issue-year, examiner, attorney, and assignee fixed effects.

Economic message: Female lead-inventor patents receive fewer citations both before and after adjusting for patent text, novelty, impact, and detailed institutional fixed effects.

Table 1
Summary statistics.
Data Source: USPTO.

Gender of lead inventor	Male			Female			Test
	N	Mean	SD	N	Mean	SD	
Panel A: Difference in forward citations							
Forward citation	289,847	23.62	62.39	232,120	19.2	55.17	F = 715.64***
Significance (standardized)	289,847	0.04	0.97	232,120	−0.05	1.03	F = 933.537***
Panel B: Difference By top decile innovations							
Breakthrough	289,847			232,120			χ^2 = 985.594***
→ No	258,237	89%		212,828	92%		
→ Yes	31,610	11%		19,292	8%		
Panel C: Difference by cooperative patent classification							
CPC section	289,847			232,120			χ^2 = 3860.740***
→ Chemistry	29,895	10%		28,058	12%		
→ Electricity	64,772	22%		60,268	26%		
→ Fixed Constructions	9,967	3%		5,544	2%		
→ Human Necessities	36,276	13%		29,996	13%		
→ Mechanical Engineering	27,683	10%		15,752	7%		
→ Performing Operations	52,116	18%		33,770	15%		
→ Physics	65,775	23%		56,473	24%		
→ Textiles	3,333	1%		2,242	1%		

This table provides summary statistics on patents and citations. The sample covers patents issued from 1976 through 2015. Panel A presents a two-way table of forward citations by gender. Panel B presents a two-way table of patents in the top decile by gender. Panel C presents a two-way table of patents by their cooperative patent classification (CPC). ***, **, * denote significance at the 1%, 5%, and 10% level, respectively.

Table 1: Summary statistics by lead-inventor gender and CPC section.

Table 2
Lead female inventors.
Data source: USPTO.

Panel A: No adjustment					
	Forward citations				
	(1)	(2)	(3)	(4)	(5)
Lead female inventor	−3.294***	−2.985***	−2.877***	−2.44***	−1.51***
	(0.411)	(0.396)	(0.381)	(0.33)	(0.22)
Art group FE	Yes	Yes	Yes	Yes	Yes
Patent issue year FE	No	Yes	Yes	Yes	Yes
Examiner FE	No	No	Yes	Yes	Yes
Attorney FE	No	No	No	Yes	Yes
Assignee FE	No	No	No	No	Yes
Observations	521 967	521 967	521 967	521 967	521 967
R^2	0.096	0.106	0.166	0.265	0.484
Panel B: Text adjustment					
	Delta in forward citations				
	(1)	(2)	(3)	(4)	(5)
Lead female inventor	−2.251***	−2.281***	−2.331***	−2.462***	−2.468***
	(0.295)	(0.296)	(0.309)	(0.312)	(0.291)
Art group FE	Yes	Yes	Yes	Yes	Yes
Patent issue year FE	No	Yes	Yes	Yes	Yes
Examiner FE	No	No	Yes	Yes	Yes
Attorney FE	No	No	No	Yes	Yes
Assignee FE	No	No	No	No	Yes
Observations	521 967	521 967	521 967	521 967	521 967
R^2	0.033	0.034	0.079	0.170	0.350

This table estimates the relationship between the gender of the lead inventor and citations. The dependent variable is the expected forward citation for each patent. This specification updates the I-TEXT architecture to include the significance measure with the text embeddings. The sample covers patents issued from 1976 through 2015. Standard errors are clustered at the patent art unit and patent issue year level. ***, **, * denote significance at the 1%, 5%, and 10% level, respectively.

Table 2: Raw and text-adjusted citation gaps for female lead-inventor patents.

6.2 Table 3 and Figure 1: Emerging fields and the I-TEXT procedure

Read Table 3:

- Panel A shows that patents in emerging fields tend to receive more actual forward citations.
- Panel B shows that the main female lead-inventor undercitation result remains when the emerging-field indicator is included.
- The interaction between emerging fields and female lead inventors is negative in the text-adjusted estimates.
- Female lead-inventor patents in emerging fields receive an additional roughly 1.4 to 1.8 fewer citations than expected under the male benchmark.

Read Figure 1:

- Patent text, gender, novelty, impact, and citations enter the estimation procedure.
- The text encoder maps patent text into embeddings.

- The propensity network estimates whether the patent's text is associated with a female lead inventor.

- Separate male and female citation networks generate counterfactual citation estimates.

Economic message: I-TEXT is the engine that turns patent text into counterfactual citation estimates. Emerging fields do not eliminate the citation gap; if anything, the text-adjusted gap is stronger there.

Table 3
Emerging fields.
Data source: USPTO.

Panel A: No adjustment					
	Forward citations				
	(1)	(2)	(3)	(4)	(5)
Emerging field	11.24***	5.01***	3.92***	3.82***	2.444**
	(2.49)	(1.67)	(1.17)	(1.13)	(0.995)
Lead female inventor	−3.382***	−3.064***	−2.957***	−2.482***	−1.583***
	(0.407)	(0.401)	(0.378)	(0.334)	(0.254)
Emerging field × Lead female inventor	−0.128	−0.802	−0.810	−1.301	0.218
	(0.890)	(0.854)	(0.783)	(0.997)	(0.903)
Art group FE	Yes	Yes	Yes	Yes	Yes
Patent issue year FE	No	Yes	Yes	Yes	Yes
Examiner FE	No	No	Yes	Yes	Yes
Attorney FE	No	No	No	Yes	Yes
Assignee FE	No	No	No	No	Yes
Observations	494 239	494 239	494 239	494 239	494 239
R^2	0.098	0.107	0.167	0.267	0.488
Panel B: Text adjustment					
	Delta in forward citations				
	(1)	(2)	(3)	(4)	(5)
Emerging field	−0.675	0.0112	0.627	0.717	0.778*
	(0.488)	(0.4777)	(0.501)	(0.465)	(0.430)
Lead female inventor	−2.221***	−2.250***	−2.308***	−2.436***	−2.45***
	(0.294)	(0.295)	(0.308)	(0.305)	(0.29)
Emerging field × Lead female inventor	−1.755**	−1.687**	−1.588**	−1.713**	−1.448**
	(0.732)	(0.731)	(0.748)	(0.741)	(0.593)
Art group FE	Yes	Yes	Yes	Yes	Yes
Patent issue year FE	No	Yes	Yes	Yes	Yes
Examiner FE	No	No	Yes	Yes	Yes
Attorney FE	No	No	No	Yes	Yes
Assignee FE	No	No	No	No	Yes
Observations	494 239	494 239	494 239	494 239	494 239
R^2	0.033	0.034	0.079	0.171	0.352

This table studies the citations to new fields of innovation. *Emerging Field* takes the value of one if the art unit first appeared within five years of the patent being granted. Panel A uses the forward citation for each patent as its dependent variable. Panel B uses I-Text adjustment, with the dependent variable being the difference in the observed number and the expected number of citations for a patent if the lead inventor was male, as defined by Eq. (8). The sample covers patents issued from 1976 through 2015. Standard errors are clustered at the art unit and patent issue year level. ***, **, * denote significance at the 1%, 5%, and 10% level, respectively.

Table 3: Emerging fields and the text-adjusted female lead-inventor citation gap.

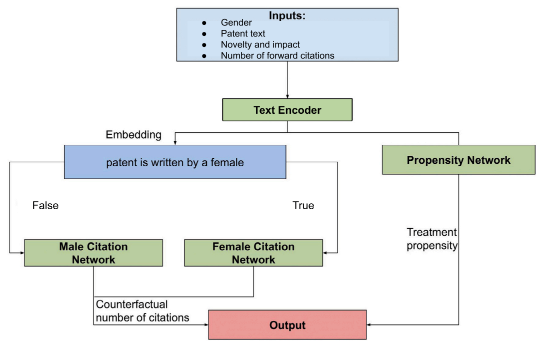


Figure 1: I-TEXT estimation procedure.

6.3 Figures 2 and 3: Citation and delta distributions

Read Figure 2:

- Panel A plots observed forward-citation distributions by gender.

- Female lead-inventor patents are more concentrated at low citation counts.

- Panel B plots I-TEXT-implied expected citation distributions.

- The expected distributions help separate observed citation outcomes from text-based patent quality.

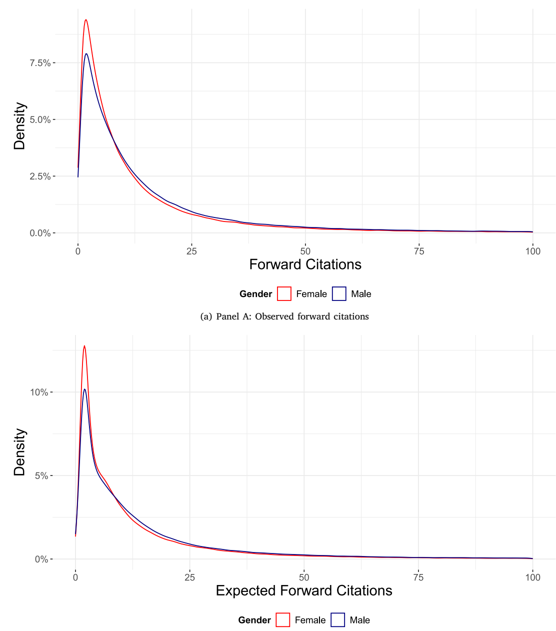
Read Figure 3:

- Δ is the difference between actual citations and expected citations under the male benchmark.

- The distribution contains both positive and negative values, so the measure is not mechanically negative.

- Female lead-inventor patents are shifted toward lower Δ values relative to male lead-inventor patents.

Economic message: The undercitation result is not simply a statement that female patents have lower raw citation counts. It is a statement about realized citations relative to what patent content and context predict.



(a) Panel A: Observed forward citations

(b) Panel B: I-TEXT implied forward citations

Figure 2: Observed and I-TEXT-implied forward-citation distributions.

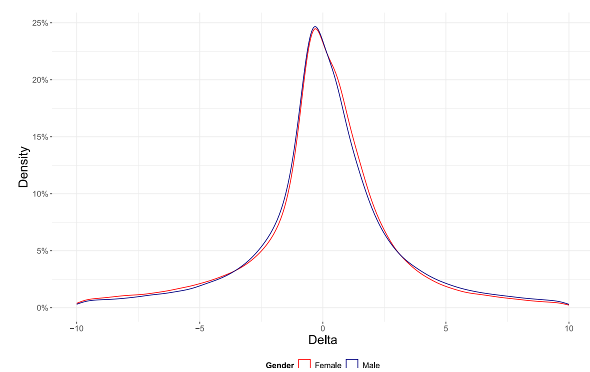


Figure 3: Distribution of Δ by lead-inventor gender.

6.4 Figures 4 and 5: Technology heterogeneity and time-series evolution

Read Figure 4:

- The figure plots adjusted Δ estimates by NBER patent subcategory.
- Most subcategory estimates are below zero, indicating broad undercitation of female lead-inventor patents.
- The magnitude varies substantially across fields.
- Some medical, genetic, computer, and other subcategories show especially large negative estimates.

Read Figure 5:

- The figure plots the gender-based citation gap by patent grant year using citations within ten years after grant.
- The gap is negative in every plotted year.
- The gap becomes most negative around the mid-to-late 1990s and then moves back toward the earlier magnitude.
- The paper does not interpret the figure as showing a clean monotonic trend over time.

Economic message: The gap is not confined to one technology class or one short period. It is broad, persistent, and heterogeneous.

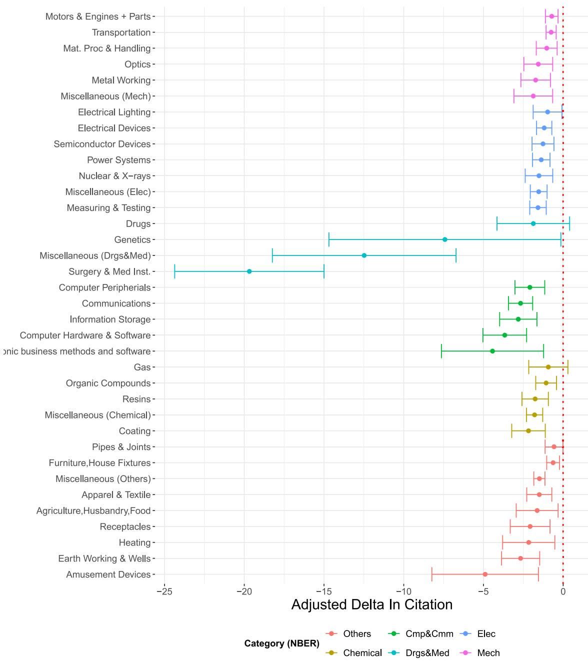


Figure 4: Text-adjusted citation gap by NBER patent subcategory.

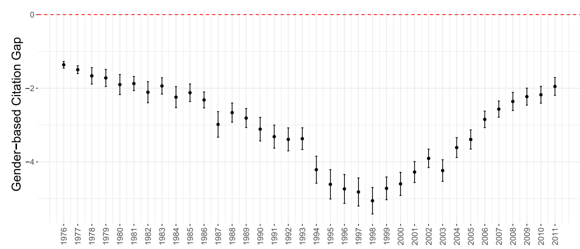


Figure 5: Evolution of the gender-based citation gap over time.

6.5 Figures 6 and 7: Who undercites and robustness

Read Figure 6:

- The largest gap comes from citations added by male lead inventors in future patent applications.
- Citations added by female lead inventors also show undercitation of prior female-led patents, but the magnitude is smaller.
- Examiner-added citation gaps are much smaller than inventor-added gaps.
- Female examiners appear closest to even-handed in the source decomposition.

Read Figure 7:

- The baseline gap is negative.
- The gap remains negative using only text data, including zero-citation patents, restricting to single-author patents, and removing self-citations.
- The placebo exercise produces a gap close to zero.

Economic message: The main result is not driven by a narrow sample definition or by the I-TEXT procedure mechanically producing negative gaps. The largest behavioral source appears to be inventor-side citation choices.

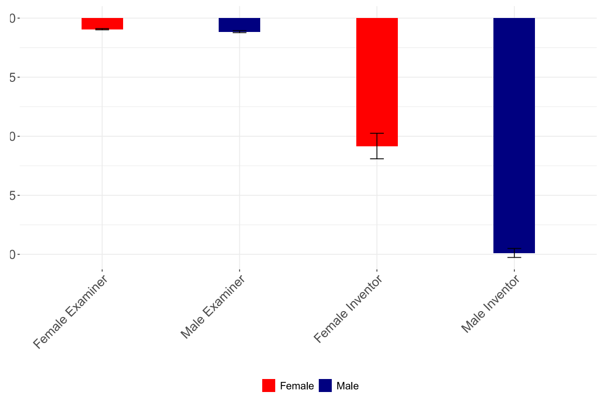


Figure 6: Gender-based citation gap by examiner- and inventor-added citations.

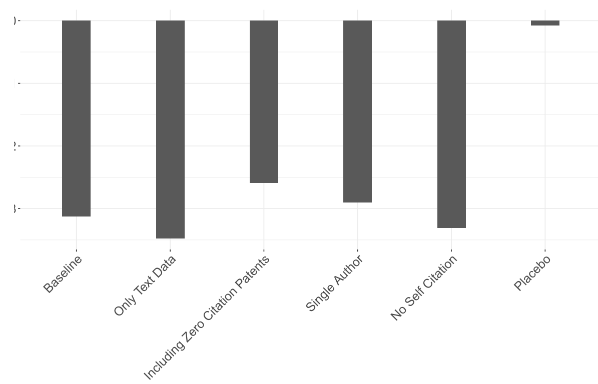


Figure 7: Gender-based citation gap across alternative samples and placebo test.

6.6 Tables 4 and 5: Examiner-added and inventor-added citations

Read Table 4:

- The dependent variable is Δ for examiner-added citations.
- Panel A focuses on citations added by female lead examiners; Panel B focuses on citations added by male lead examiners.
- Examiner-added citation gaps are economically small relative to the baseline citation gap.
- In the most saturated specifications, the examiner-added estimates are small and not statistically significant.

Read Table 5:

- The dependent variable is Δ for inventor-added citations.
- Panel A shows that female lead inventors undercite prior female lead-inventor patents modestly.
- Panel B shows that male lead inventors undercite prior female lead-inventor patents much more strongly.
- Male inventor-added citation gaps are roughly -1.6 to -1.8 citations and are statistically significant across specifications.

Economic message: The undercitation of female-led patents appears to come mainly from future inventors, especially male lead inventors, rather than from patent examiners.

Table 4
Examiner gender counterfactuals.
Data source: USPTO.

Panel A: Citation added by female lead examiner					
	Delta in forward citations				
	(1)	(2)	(3)	(4)	(5)
Lead female inventor	−0.1185***	−0.1101***	−0.1149***	−0.1374**	−0.0689
	(0.0246)	(0.0249)	(0.0393)	(0.0523)	(0.1398)
Art group FE	Yes	Yes	Yes	Yes	Yes
Patent issue year FE	No	Yes	Yes	Yes	Yes
Examiner FE	No	No	Yes	Yes	Yes
Attorney FE	No	No	No	Yes	Yes
Assignee FE	No	No	No	No	Yes
Observations	17 603	17 603	17 603	17 603	17 603
R^2	0.089	0.094	0.414	0.671	0.799
Panel B: Citation added by male lead examiner					
	Delta in forward citations				
	(1)	(2)	(3)	(4)	(5)
Lead female inventor	−0.1111***	−0.1049***	−0.1028***	−0.0842**	−0.0679
	(0.0276)	(0.0281)	(0.0307)	(0.0331)	(0.0415)
Art group FE	Yes	Yes	Yes	Yes	Yes
Patent issue year FE	No	Yes	Yes	Yes	Yes
Examiner FE	No	No	Yes	Yes	Yes
Attorney FE	No	No	No	Yes	Yes
Assignee FE	No	No	No	No	Yes
Observations	72 505	72 505	72 505	72 505	72 505
R^2	0.030	0.033	0.204	0.400	0.561

This table studies the source of examiner-added citations for male inventors. The dependent variable is the difference in forward citations. The dependent variable is *Delta*, the difference in the observed number and the expected number of citations for a patent if the lead author was male, as defined by Eq. (8). Panel A uses the difference in forward citations that were added by female lead examiners as its dependent variable. Panel B uses the difference in forward citations that were added by male lead examiners as its dependent variable. The sample covers patents issued from 1976 through 2015. Note, the source of citations is only available following the start of 2001. The sample covers patents issued from 1976 through 2015. Standard errors are clustered at the art unit and patent issue year level. ***, **, * denote significance at the 1%, 5%, and 10% level, respectively.

Table 4: Examiner-added citations by examiner gender.

Table 5
Inventor-added citations.
Data source: USPTO.

Panel A: Citation added by female lead inventors					
	Delta in forward citations				
	(1)	(2)	(3)	(4)	(5)
Lead female inventor	−0.384	−0.479*	−0.496	−0.301 701***	−0.515 619***
	(0.279)	(0.250)	(0.739)	(0.000 001)	(0.000 001)
Art group FE	Yes	Yes	Yes	Yes	Yes
Patent issue year FE	No	Yes	Yes	Yes	Yes
Examiner FE	No	No	Yes	Yes	Yes
Attorney FE	No	No	No	Yes	Yes
Assignee FE	No	No	No	No	Yes
Observations	4727	4727	4727	4727	4727
R^2	0.279	0.294	0.761	0.923	0.954
Panel B: Citation added by male lead inventors					
	Delta in forward citations				
	(1)	(2)	(3)	(4)	(5)
Lead female inventor	−1.627***	−1.629***	−1.649***	−1.760***	−1.696***
	(0.238)	(0.238)	(0.247)	(0.237)	(0.257)
Art group FE	Yes	Yes	Yes	Yes	Yes
Patent issue year FE	No	Yes	Yes	Yes	Yes
Examiner FE	No	No	Yes	Yes	Yes
Attorney FE	No	No	No	Yes	Yes
Assignee FE	No	No	No	No	Yes
Observations	266 015	266 015	266 015	266 015	266 015
R^2	0.021	0.022	0.078	0.214	0.355

This table studies the source of inventor-added citations. The dependent variable is the difference in forward citations. The dependent variable is *Delta*, the difference in the observed number and the expected number of citations for a patent if the lead inventor was male, as defined by Eq. (8). Panel A uses the difference in forward citations that were added by female lead inventors as its dependent variable. Panel B uses the difference in forward citations that were added by male lead inventors as its dependent variable. The sample covers patents issued from 1976 through 2015. Note, the source of citations is only available following the start of 2001. Standard errors are clustered at the art unit and patent issue year level. ***, **, * denote significance at the 1%, 5%, and 10% level, respectively.

Table 5: Inventor-added citations by inventor gender.

6.7 Table 6: Economic value of patents

Read:

- Table 6 uses the Kogan et al. market-based dollar value of patents.
- Panel A relates patent value to the female lead-inventor indicator and fixed effects.
- The negative coefficient in simpler specifications becomes statistically indistinguishable from zero once richer fixed effects are included.
- Panel B adds actual citations, expected male-benchmark citations, novelty, and impact.
- The female lead-inventor coefficient remains small and statistically insignificant.
- Expected male-benchmark citations and novelty load positively in many specifications, while actual citations do not consistently explain market value once the richer controls are included.

Economic message: The market does not appear to assign lower value to female-led patents after controls. This supports the interpretation that the citation gap reflects under-recognition in citations rather than lower economic value.

Table 6
Value of patent.
Data source: USPTO.

Panel A: Without controls					
	Dollar value				
	(1)	(2)	(3)	(4)	(5)
Lead female inventor	−0.41**	−0.26*	−0.27*	−0.15	−0.12
	(0.17)	(0.15)	(0.14)	(0.11)	(0.15)
Art group FE	Yes	Yes	Yes	Yes	Yes
Patent issue year FE	No	Yes	Yes	Yes	Yes
Examiner FE	No	No	Yes	Yes	Yes
Attorney FE	No	No	No	Yes	Yes
Assignee FE	No	No	No	No	Yes
Observations	191 086	191 086	191 086	191 086	191 086
R^2	0.090	0.115	0.188	0.450	0.593

Panel B: With citation and novelty controls					
	Dollar value				
	(1)	(2)	(3)	(4)	(5)
Lead female inventor	−0.21	−0.17	−0.19	−0.12	−0.12
	(0.15)	(0.14)	(0.14)	(0.11)	(0.15)
log(Actual Citations)	0.17	0.061	0.095	0.029	−0.11
	(0.22)	(0.173)	(0.178)	(0.171)	(0.11)
log(Expected Male)	1.84***	1.55***	1.46***	0.80***	0.38**
	(0.46)	(0.37)	(0.39)	(0.29)	(0.16)
log(novelty)	2.62***	1.01**	1.28**	0.92**	0.39*
	(0.86)	(0.42)	(0.51)	(0.41)	(0.21)
log(impact)	1.10	−0.27	−0.0062	0.24	0.29
	(0.74)	(0.27)	(0.3482)	(0.32)	(0.22)
Art group FE	Yes	Yes	Yes	Yes	Yes
Patent issue year FE	No	Yes	Yes	Yes	Yes
Examiner FE	No	No	Yes	Yes	Yes
Attorney FE	No	No	No	Yes	Yes
Assignee FE	No	No	No	No	Yes
Observations	191 075	191 075	191 075	191 075	191 075
R^2	0.098	0.119	0.191	0.451	0.593

This table presents the relationship between the gender of the lead inventor and the real dollar value of the granted patent. The dependent variable is deflated to 1982 (million) dollars using the CPI, as calculated in Kogan et al. (2017). The sample covers patents with a positive number of citations, granted from 1976 through 2015. Standard errors are clustered at the art unit and patent issue year level. ***, **, * denote significance at the 1%, 5%, and 10% level, respectively.

Table 6: Lead inventor gender and market-based patent value.

7 Short summary

- The paper studies whether patents with female lead inventors are under-cited relative to comparable male lead-inventor patents.
- It uses USPTO utility patents granted from 1976 through 2015.
- The main empirical challenge is that citations are both the outcome and the usual proxy for quality.
- The paper introduces I-TEXT, which uses patent text, novelty, and impact to estimate counterfactual citation counts.
- Longformer embeddings represent the full patent text, while novelty and impact capture the

patent's position relative to prior and future patents.

- The key outcome is Δ: actual citations minus expected citations under a male lead-inventor benchmark.
- Female lead-inventor patents receive about 2.5 fewer citations than the same patents are predicted to receive if the lead inventor were male.
- The undercitation pattern is broad across technology areas and persistent over time.
- Emerging fields show an especially strong text-adjusted female undercitation gap.
- Male lead inventors are the largest contributors to the undercitation of past female-led patents.
- Female lead inventors also undercite female-led patents, but by less.
- Examiner-added citation gaps are much smaller; female examiners appear closest to even-handed.
- Robustness checks using alternative samples, zero-citation patents, single-inventor patents, no self-citations, alternative embeddings, and placebo tests support the main conclusion.
- Market-based patent values do not show a comparable gender gap after controls.
- The paper implies that forward citations may be a biased measure of patent quality when gender is relevant.

8 Conclusion

8.1 Core conclusion

Hochberg, Kakhbod, Li, and Sachdeva show that patents with female lead inventors are under-cited relative to what the same patents would be expected to receive if the lead inventor were male. The result remains after adjusting for patent text, novelty, impact, technology area, issue year, examiner, attorney, and assignee fixed effects.

8.2 Economic intuition

Patent citations are a form of recognition. If inventors and examiners fail to cite female-led patents at the same rate as comparable male-led patents, then citations will understate the contribution of female inventors. This can matter because citations influence perceptions of patent quality, inventor ability, and the importance of technological contributions.

8.3 Incremental contribution revisited

The paper's key contribution is to separate patent quality from patent citations using machine learning on text. Prior work could show that female inventors receive fewer citations, but this paper provides a way to ask whether the same patent would receive different citation credit under a different lead-inventor gender.

8.4 Implications

The findings have two main implications. First, undercitation may reduce recognition for female inventors and potentially affect career outcomes, startup financing, promotion decisions, and incentives to enter innovation-intensive fields. Second, researchers using forward citations as a proxy for patent quality should account for the possibility that citations are systematically biased.

8.5 Limitations

The analysis is not a randomized experiment. The I-TEXT approach depends on the assumption that patent text, novelty, and impact capture the relevant dimensions of patent quality. The paper also cannot fully identify why undercitation occurs. The observed pattern could reflect discrimination, professional networks, visibility, information flow, or other structural mechanisms.

8.6 Takeaway

The central takeaway is that citations measure recognition, not just quality. Patents with female lead inventors appear to receive less citation recognition than their content and economic value would justify, which means citation-based measures can understate female inventors' contributions to innovation.